## Chapter 9

**Squeeze Theorem for Sequences.** Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ with $a_n \leq b_n \leq c_n$ for all integers $n$ greater than some index $N$. If $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = L$, then $\lim_{n\to\infty} c_n = L$.

**Growth Rate of Sequences.** The following sequences are ordered according to increasing growth rates as $n \to \infty$: $\{\ln n\} \ll \{n^p\} \ll \{n^p \ln n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\}$. The ordering applies for positive real numbers $p, r$, and $s$ and $b > 1$.

**Geometric Series.** Let $r, a \in \mathbb{R}$. If $|r| < 1$, then $\sum\limits_{k=0}^{\infty} ar^k = \frac{a}{1-r}$. If $|r| \geq 1$, then the series diverges.

**Divergence Test.** If $\sum a_k$ converges, then $\lim\limits_{n\to\infty} a_k = 0$. Equivalently, if $\lim\limits_{n\to\infty} a_k \neq 0$, then the series diverges.

**Integral Test.** Suppose $f$ is a continuous, positive, decreasing function for $x \geq 1$ and let $a_k = f(k)$ for $k = 1, 2, 3, \ldots$. Then

$$\sum_{k=1}^{\infty} a_k \text{ and } \int_1^{\infty} f(x)\, dx$$

either both converge or both diverge. In the case of convergence, the value of the integral is *not*, in general, equal to the value of the series.

**Ratio Test.** Let $\sum a_k$ be an infinite series with positive terms and let $r = \lim\limits_{n\to\infty} \dfrac{a_{k+1}}{a_k}$.

**1.** If $0 \leq r < 1$ then the series converges.

**2.** If $r > 1$ (including $r = \infty$) then the series diverges.

**3.** If $r = 1$ then the test is inconclusive.

**Root Test.** Let $\sum a_k$ be an infinite series with positive terms and let $\rho = \lim\limits_{n\to\infty} \sqrt[k]{a_k}$.

**1.** If $0 \leq \rho < 1$ then the series converges.

**2.** If $\rho > 1$ (including $r = \infty$) then the series diverges.

**3.** If $\rho = 1$ then the test is inconclusive.

**Comparison Test.** Let $\sum a_k$ and $\sum b_k$ be series with positive terms.

**1.** If $0 < a_k \leq b_k$ and $\sum b_k$ converges, then $\sum a_k$ converges.

**2.** If $0 < b_k \leq a_k$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

**The Limit Comparison Test.** Let $\sum a_k$ and $\sum b_k$ be series with positive terms and let $\lim\limits_{k\to\infty} \dfrac{a_k}{b_k} = L$.

**1.** If $0 < L < \infty$ then $\sum a_k$ and $\sum b_k$ either both converge or diverge.

**2.** If $L = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.

**3.** If $L = \infty$ and $\sum b_k$diverges, then $\sum a_k$ diverges.

## Chapter 10

**Taylor's Polynomials.** Let $f$ be a function with $f', f'', \dots f^{(n)}$ defined ay $a$. The ***n*th-order Taylor Polynomial** for $f$ with its **center** at $a$m denoted $p_n$, has the property that it matches $f$ in value, slope, and all derivatives up to the $n$th derivative at $a$; that is,

$$p_n(a) = f(a) \text{ and } p_n'(a) = f'(a), \dots, p_n^{(n)}(a) = f^{(n)}(a).$$

The $n$th-order Taylor polynomial centered at $a$ is

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

More compactly,

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \text{ for } k = 0, 1, 2, \dots, n.$$

## Chapter 11

**Slope of a Tangent Line.** Let $f$ be differentiable function at $\theta_0$. The slope of the line tangent to the curve $r = f(\theta)$ at the point $(f(\theta), \theta_0)$ is

$$\frac{dy}{dx} = \frac{f'(\theta_0)\sin\theta_0 + f(\theta_0)\cos\theta_0}{f'(\theta_0)\cos\theta_0 - f(\theta_0)\sin\theta_0},$$

provided the denominator is nonzero at the point. At angles $\theta_0$ for which $f(\theta_0) = 0$ and $f'(\theta_0) \neq 0$, the tangent line is $\theta = \theta_0$ with slope $\tan\theta_0$.

**Area of Regions in Polar Coordinates.** Let $R$ be the region bounded by the graph of $r = f(\theta)$ and the origin, between $\theta = \alpha$ and $\theta = \beta$, where $f$ is continuous and $f(\theta) \geq g(\theta) \geq 0$ on $[\alpha, \beta]$. The area of $R$ is

$$\int_\alpha^\beta \frac{1}{2} f(\theta)^2 d\theta.$$

## Helpful formulas

1. $\frac{d}{dx}\sin ax = a\cos ax$

2. $\frac{d}{dx}\cos ax = -a\sin ax$

3. $\int \cos ax\, dx = \frac{1}{a}\sin ax + C$

4. $\int \sin ax\, dx = -\frac{1}{a}\cos ax + C$

5. $\int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$

6. $\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$